

ImageLab Progress Summary Report

Project: ImageLab

Reporting Period: Week 2

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Week 2 focused on transforming ImageLab from a simple image-upload application into a fully functional asynchronous processing system. The main goal was to allow image uploads to be accepted immediately while processing occurs in the background. By the end of the week, the complete backend workflow was operational, including job creation, worker execution, status tracking, variant generation, and error handling.

The most significant enhancement was the implementation of the async acceptance contract. The `POST /v1/images` endpoint now creates a queued job after storing the uploaded image and returns a `202 Accepted` response instead of `201 Created`. The response includes the image ID, job ID, status, status URL, and `Location` header, giving clients everything needed to track progress while separating work acceptance from work completion.

To support monitoring, the `GET /v1/jobs/{id}` endpoint was implemented. It returns the current job status, timestamps, and, once processing is complete, the generated image variants and their URLs.

A background worker was also developed. Jobs are claimed transactionally using `SELECT ... FOR UPDATE SKIP LOCKED`, ensuring only one worker processes a job at a time. The worker generates the required image variants using Go's standard image libraries and marks the job as complete only after all variants have been successfully created and recorded. If processing fails, a client-safe error message is stored.

Additional functionality was added through the `GET /v1/images/{id}/variants/{name}` endpoint, which safely serves generated variants. Variant names are validated against an allowlist before file access, preventing path manipulation and restricting requests to expected resources.

Testing identified and resolved several issues. A thumbnail-cropping bug affecting non-square images was fixed, logging was improved by preventing empty queues from being reported as errors, and a frontend issue that allowed duplicate image submissions was corrected by keeping the processing button disabled until a new image is selected.

By the end of Week 2, the backend async processing system was fully operational and met project requirements. The remaining work focuses on the frontend, including polling, live status updates, result rendering, retry handling, and cancellation support, all planned for Week 3. The next phase will connect the frontend to the completed backend, allowing users to track jobs and view generated image variants in real time.